

Please check that this question paper contains 09 questions and 03 printed pages within first ten minutes.

MORNING

18 JUN 2025

[Total No. of Questions: 09]

[Total No. of Pages: 03]

Uni. Roll No.

Program: B.Tech. (Batch 2018 onward)

Semester: 3

Name of Subject: Mathematics-III

Subject Code: BSCS-101

Paper ID: 16013

Scientific calculator is Allowed

Normal Distribution table is allowed

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory
- 2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
- 3) Any missing data may be assumed appropriately

Part – A

[Marks: 02 each]

Q1

- a) Separate $\cos(x + iy)$ into real and imaginary parts.
- b) Write algorithm for straight line fit of a curve.
- c) Check the correctness of statement “Mean of a Binomial Distribution is 3 and variance is 5”.
- d) Evaluate $\oint_C \frac{e^{-z}}{z+1} dz$, where C is the circle $|z|=2$.
- e) Solve the system of equations $x + y = 5$, $2x - y = 4$ using Gauss-Jordan method.
- f) The number of emergency admission each day to hospital is found to have Poisson distribution with mean 4. Find the probability that on a particular day there will be no emergency admission. ($e^{-4} = 0.0183$)

Part – B**MORNING****18 JUN 2025****[Marks: 04 each]**

Q2. Determine the analytic function $f(z)$ in terms of z whose real part is $e^x(x \cos y - y \sin y)$.

Q3. Fit a straight line to the following data:

x	1	2	3	4	5
y	5	7	9	10	11

Q4. Solve the system of equations $x + 2y + z = 3$, $2x + 3y + 3z = 10$, $3x - y + 2z = 13$ using Gauss elimination method.

Q5. Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in the regions (i) $1 < |z| < 3$ (ii) $|z| > 3$

Q6. The probability distribution of a variable X is given by

X	0	1	2	3	4	5	6
$P(X)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$

(i) Find k (ii) Evaluate $P(X < 4)$ and $P(X \geq 5)$.

Q7. Random samples of 400 men and 600 women asked whether they would like to have flyover near their residence. If 200 men and 325 women were in favour of proposal. Test the hypothesis that proportions of men and women in favour of the proposal are same at 5% level of significance. (tabulated value = 1.96)

Part – C**[Marks: 12 each]**

Q8. Evaluate the integral $\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta}$, where $a > |b|$ using contour integration.

OR

If $f(z)$ is an analytic function of z , prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4|f'(z)|^2$.

Q9. A set of 5 similar coins is tossed 320 times and the result is

No. of heads	0	1	2	3	4	5
Frequencies	6	27	72	112	71	32

MORNING

18 JUN 2025

Test the hypothesis that the data follows Binomial distribution.

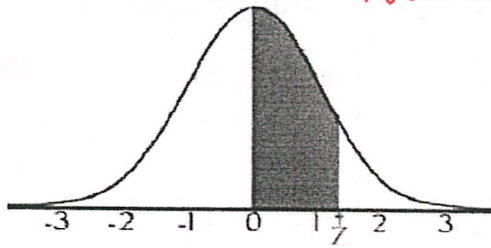
(For $\nu = 5$, $\chi^2_{0.05} = 11.07$)

OR

A sample of 100 dry battery cells tested to find the length of life produced the following results: $\mu = 12$ hours and $\sigma = 3$ hours. Assume that the data to be normally distributed, what percentage of battery cells are expected to have life

- (a) more than 15 hours
- (b) less than 6 hours
- (c) between 10 and 14 hours.

18 JUN 2025



STANDARD NORMAL TABLE (Z)

Entries in the table give the area under the curve between the mean and z standard deviations above the mean. For example, for $z = 1.25$ the area under the curve between the mean (0) and z is 0.3944.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998